

LAB 4: WORKING WITH THREADS ON LINUX

I. Objectives

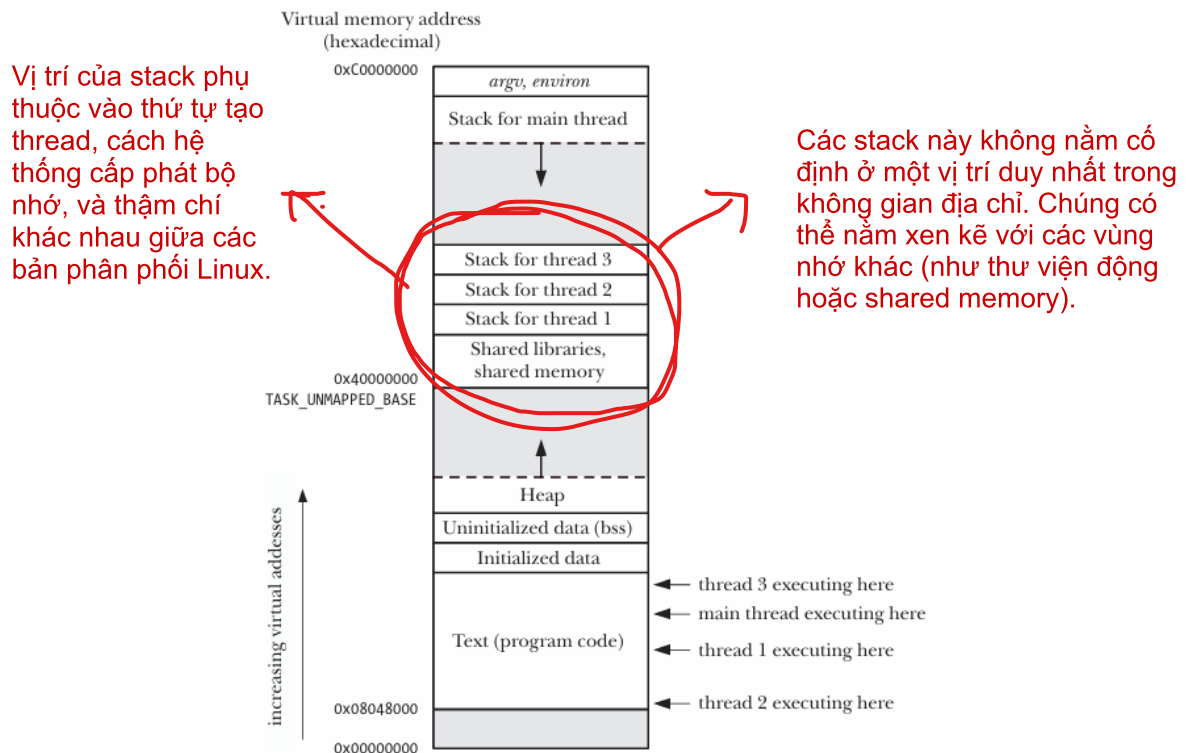
- Students understand the concept of thread, differences between thread and process, thread synchronization with mutex & condition variable and basic system calls when working with threads.
- Students can using system calls and library functions for required tasks on threads handling.

II. Basic theory

1. Overview of thread

1.1 Introduction to thread

Like processes, threads are a mechanism that permits an application to perform multiple tasks concurrently. A **single process** can **contain multiple threads**. All of these threads are **independently executing the same program**, and they all **share the same global memory**, including the initialized data, uninitialized data, and heap segments.



đồng thời

The threads in a process can execute **concurrently**. On a multiprocessor system, **multiple threads** can execute parallel. If one thread is blocked on I/O, other threads are still eligible to execute.

đan xen

In particular, the location of the per-thread stacks may be **intermingled** with **shared libraries and shared memory regions**, depending on the **order** in which threads are created, **shared libraries loaded**, and **shared memory regions attached**. Furthermore, the location of the per-thread stacks can **vary depending on the Linux distribution**.

Threads have some advantages over using multiple processes in applications that need to execute concurrently:

- **Sharing information between threads** is easy and fast. It is just a **matter of copying data into shared (global or heap) variables**.
- **Thread creation is faster** than process creation—typically, **ten times faster** or better. Thread creation is faster because many of the attributes that must be duplicated in a child created by `fork()` are instead shared between threads. In particular, copy-on-write duplication of pages of memory is not required, nor is duplication of page tables.

Besides **global memory**, threads also share a number of other attributes (i.e., **these attributes are global to a process**, rather than specific to a thread). These attributes include the following:

- process ID and parent process ID;
- process group ID and session ID;
- controlling terminal;
- process credentials (user and group IDs);
- open file descriptors;
- record locks created using `fcntl()`;
- signal dispositions;
- file system-related information: umask, current working directory, and root directory;
- interval timers (`setitimer()`) and POSIX timers (`timer_create()`);
- System V semaphore undo (`semadj`) values (Section 47.8);

Các thuộc tính chung
cho toàn bộ process
,mọi thread đều thấy và
dùng chung

- resource limits;
- CPU time consumed (as returned by times());
- resources consumed (as returned by getrusage()); and
- nice value (set by setpriority() and nice())

Among the **attributes that are distinct for each thread** are the following:

- thread ID;
- signal mask;
- thread-specific data;
- alternate signal stack (sigaltstack());
- the errno variable;
- floating-point environment (see fenv(3));
- realtime scheduling policy and priority;
- CPU affinity (Linux-specific,);
- capabilities (Linux-specific); and
- stack (local variables and function call linkage information)

Các thuộc tính riêng cho từng thread

1.2 Thread creation

When a **program is started**, the resulting process **consists of a single thread**, called the **initial or main thread**. In this section, we look at how to **create additional threads**.

The **pthread_create()** function creates a new thread.

```
#include <pthread.h>
```

```
int pthread_create(pthread_t *thread, const pthread_attr_t *attr, void  
*(*start)(void *), void *arg);
```

Returns 0 on success, or a **positive error number** on error

Parameters:

- cũng có:
- hàm execute của thread ~ handler trong IPD
 - system call để đăng ký: pthread_create() ~ signal()

việc tạo thread cũng khá tương đồng khi thực hành IPC trong process

con trỏ trỏ tới biến kiểu pthread, lưu ID của thread mới

- **thread**: A pointer to a pthread_t variable, where the new thread's ID is stored.
- **attr**: A pointer to a pthread_attr_t object that specifies the thread's attributes (can be NULL to use default attributes).
- **start**: A pointer to the function that the new thread will execute.
- **arg**: The argument passed to the start function.

con trỏ tới hàm thực thi của thread mới

con trỏ tới vùng chứa lưu tham số của hàm đó

Work:

- The new thread begins execution at the start() function, receiving the arg parameter.
- The thread that calls pthread_create() continues executing without waiting for the new thread to finish.
- The arg parameter is of type void *, allowing it to store a pointer to any data type (typically a pointer to a global or heap variable).
- If multiple arguments need to be passed, arg can be a pointer to a struct containing those arguments.

object

Considerations:

- Be cautious when using an integer (int) as the return value of the thread function, as it may conflict with PTHREAD_CANCELED (a special value returned when a thread is canceled).
- The new thread's ID is stored in pthread_t, but its initialization is not guaranteed before the new thread starts running.
- If the new thread needs to obtain its own ID, it must call pthread_self().
- No guaranteed scheduling order exists between threads, so if execution order matters, synchronization mechanisms must be used.

mutex, semaphore, pthread_cond_t, pthread_barrier_t

1.3 Thread termination

A thread can terminate in one of the following ways:

- Returning from the start function: The thread's **start function completes** and **returns a value**.
- Calling **pthread_exit()**: The thread **explicitly** terminates using **pthread_exit()**.
- Being **canceled using pthread_cancel()**: Another thread cancels it
- **Process-wide termination:**
 - Any thread calls **exit()**, terminating the entire process. *đây là system calls để kết thúc tiến trình ko phải thread dùng pthread_exit() để kết thúc thread*
 - The **main thread returns from main()**, which **terminates all threads immediately**. *thread chính return là terminate tất cả các thread còn lại*

```
#include <pthread.h>
```

```
void pthread_exit(void *retval);
```

Khá tương đồng với exit() và wait() trong process system calls

- Terminates the calling thread.
- The retval parameter allows the thread to specify a return value.
- Another thread can retrieve this return value using **pthread_join()**.

Considerations:

- Calling **pthread_exit()** is similar to **returning from the thread's start function** but **can be done from any function within the thread's execution**.
- The **retval should not be stored on the thread's stack**, as the stack's memory becomes **invalid upon thread termination** and might be reused by a new thread.
- If the **main thread calls pthread_exit()**, the **process remains running** as long as **other threads are still executing**. *main thread mà gọi thì chỉ có main thread dừng thôi*
- If the main thread calls **exit()** or returns from **main()**, all threads terminate immediately.

1.4 Thread ID

Each thread within a process is uniquely identified by a **thread ID**. This ID is returned to the caller of **pthread_create()**, and a thread can obtain its own ID using **pthread_self()**. *ghi vào tham số tid*

←→ getpid()

Khi một thread kết thúc, stack của nó sẽ bị giải phóng.

Nếu bạn trả về một con trỏ tới biến nằm trên stack của thread đó, thì vùng nhớ đó không còn hợp lệ nữa → dễ gây lỗi hoặc dữ liệu rác

```
#include <pthread.h>
```

```
pthread_t pthread_self(void);
```

Returns the thread ID of the calling thread

Uses of Thread IDs

- **Thread management:** Many Pthreads functions use thread IDs to identify the target thread, such as:
 - pthread_join()
 - pthread_detach()
 - pthread_cancel()
 - pthread_kill()
- **Data structure tagging:** Thread IDs can be used to mark dynamic data structures, helping to:
 - Track which thread created or owns a resource.
 - Identify which thread should later process that data.

Comparing Thread IDs

To check if two thread IDs are the same, use `pthread_equal()`:

```
#include <pthread.h>
```

```
int pthread_equal(pthread_t t1, pthread_t t2);
```

Returns a nonzero value if t1 and t2 are the same, otherwise returns 0

Example: Checking if the calling thread matches a stored thread ID

```
if (pthread_equal(tid, pthread_self()))  
    printf("tid matches self\n");
```

Use `pthread_equal()` instead of “==”:

- `pthread_t` is an opaque data type and its representation can vary across systems. kiểu dữ liệu mờ đục, có thể khác trên các hệ điều hành khác
- On Linux, `pthread_t` is defined as an unsigned long, but on other systems, it may be a pointer or structure. size = 4 bytes

Đây là một chuẩn do The Open Group đưa ra, nhằm thống nhất cách các hệ điều hành kiểu UNIX (Linux, BSD, Solaris, v.v.) cung cấp API.

- **SUSv3** does not guarantee that **pthread_t** is a scalar type, so direct comparisons (==) may not work portably.

Scalar type = kiểu dữ liệu đơn giản, có thể so sánh trực tiếp bằng toán tử ==

Thread IDs in Linux vs Other Systems:

- On **Linux**, thread IDs are **unique** across processes.
- However, on **other systems**, thread IDs **may not be unique** outside their own process.
- Thread IDs may be **reused** after:
 - A thread is joined using **pthread_join()**.
 - A detached thread terminates.
- Linux provides a system call (**gettid()**) that returns a kernel-assigned thread ID, which is similar to a process ID.

Là system call của Linux, trả về Thread ID do kernel gán

1.5 Joining with a terminated thread

pthread_self() thì trả về kiểu dữ liệu pthread_t nội bộ thư viện pthread

The **pthread_join()** function allows a thread to wait for another thread to terminate. If the specified thread has already finished, **pthread_join()** returns immediately. Join thread = chờ thread khác kết thúc để lấy kết quả và dọn dẹp tài nguyên.

```
#include <pthread.h>
```

```
int pthread_join(pthread_t thread, void **retval);
```

Returns 0 on success or a positive error number on failure.

Parameters:

- **thread**: The thread ID of the thread to join.
- **retval**: A **pointer** to **store the return value** of the terminated thread (as passed in **pthread_exit()** or **the return value of the thread function**). Can be **NULL** if not needed.

Considerations:

- If **pthread_join()** is called on a thread that has **already been joined**, **unpredictable behavior can occur**. The ID might be reused by another thread, leading to unintended joins.

- Joining is **necessary for non-detached threads**. If a thread is **not joined**, it becomes **a zombie thread**, wasting system resources. **If too many zombie threads accumulate, no new threads can be created.**

Comparison: `pthread_join()` vs `waitpid()`

`pthread_join()` for threads is **similar to `waitpid()` for processes** but with important differences:

- Threads are **peers**: **bất cứ thread nào, TRONG CÙNG 1 PROCESS cũng có thể tạo thêm thread, không nhất thiết phải là main thread**
 - Any thread in a process **can call `pthread_join()` to join any other thread in the process.** **thread A tạo thread B, thread B tạo thread C**
 - Example: If Thread A creates Thread B, which creates Thread C, then: **Thread A can join with C, or Thread C can join with A.**
 - This is **different from processes**, where **only the parent process can `wait()` on its child.** **chỉ có tiến trình cha đợi con thôi, các tiến trình khác không đợi**
- No "join any thread" option:
 - Unlike processes (where `waitpid(-1, &status, options)` can **wait for any child process**), `pthread_join()` only joins with a specific thread ID. **process cha thì có thể đợi bất cứ child nào terminate nhưng thread phải dùng ID để đợi (join)**
 - There is also **no non-blocking join** (unlike `waitpid(WNOHANG)`), which checks without blocking). **Workaround:** Similar behavior can be achieved **using condition variables.** **vừa đợi vừa execute**

Sau khi detach, không thể join và không lấy được giá trị trả về.

1.6 Detaching a thread **detached → hệ thống tự giải phóng khi thread kết thúc.**

By **default**, a thread is **joinable**, meaning that **when it terminates, another thread can retrieve its return status** using `pthread_join()`. However, **in some cases, we don't need the thread's return status** and simply **want the system to automatically clean up the thread** when it terminates. In such cases, we can **mark the thread as detached** using `pthread_detach()`.

```
#include <pthread.h>
```

```
int pthread_detach(pthread_t thread);
```

```
Returns 0 on success or a positive error number on failure
```


Parameters:

- **thread**: The thread ID of the thread to join.

Considerations:

- Once a thread is detached, it cannot be joined using `pthread_join()`, and its return status cannot be retrieved.
- A detached thread cannot be made joinable again. *vẫn có thể terminate 1 thread đã detach*
- Detaching a thread does not prevent it from being terminated if:
 - Another thread calls `exit()`. *Trong process đó, thread đầu tiên chính là main thread, nó bắt đầu thực thi từ hàm main().*
 - The main thread returns from `main()`. *terminate cả process luôn*
 - From the main thread, you can call `pthread_create()` to create other threads. *Từ main thread, bạn có thể gọi pthread_create() để tạo thêm các thread khác*
- In these cases, all threads in the process (including detached threads) terminate immediately.
- `pthread_detach()` only controls what happens after a thread terminates, but it does not affect how or when the thread terminates. *không có giá trị trả về từ thread đó nữa*

Example:

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>

void *threadFunc(void *arg) {
    printf("Hello from thread!\n");
    return NULL;
}

int main() {
    pthread_t thread;

    // Create a thread
    pthread_create(&thread, NULL, threadFunc, NULL);

    printf("Hello from main!\n");
```

```
// Wait for the thread to finish
pthread_join(thread, NULL);

return 0;
}
```

nhưng mà anh đã thử hỏi chatgpt xem là có áp dụng với version GCC nào cụ thể không, thì nó trả lời là áp dụng cho tất cả các version.

nhưng trong repo anh đã cấu hình để compile tự động mà không có -pthread, vẫn chạy ok

Note: GCC does not automatically link the pthread library so you need to explicitly link it (using -pthread flag) when compile your program.

gcc -pthread example.c -o example

GCC mặc định chỉ biên dịch và liên kết với thư viện chuẩn C (libc). Nó không tự động thêm pthread.

1.7 Thread attributes

When creating a thread using pthread_create(), the attr argument (of type pthread_attr_t) can be used to customize the thread's attributes.

Although we won't go into full details, these attributes include:

- **Stack location and size:** Specifies where and how large the thread's stack should be.
- **Scheduling policy and priority:** Similar to process scheduling policies.
- **Joinable or detached status:** Determines whether the thread will need to be joined later (joinable) or if it will clean up itself automatically (detached).

Example: Creating a Detached Thread

Instead of detaching a thread after creation using pthread_detach(), a thread can be created already detached by setting the appropriate attribute in pthread_attr_t.

tức là thay vì tạo thread xong thì mới detach thì tạo 1 thread có attribute là detach luôn

Process:

- Initialize a pthread_attr_t structure with default values.
- Modify the attribute to specify that the thread should be detached.
- Use pthread_create() while passing the modified attributes.
- Once the thread is created, destroy the pthread_attr_t object since it is no longer needed.

This method allows better control over thread behavior at creation time rather than modifying attributes later.

2. Thread synchronization with mutex

cấu hình stack của thread

chính sách lập lịch và độ ưu tiên của thread

trạng thái join

2.1 Introduction to mutex

To avoid the problems that can occur when threads try to update a shared variable, we must use a mutex (short for mutual exclusion) to ensure that only one thread at a time can access the variable. More generally, mutexes can be used to ensure atomic access to any shared resource, but protecting shared variables is the most common use.

A mutex has two states: **locked** and **unlocked**. At any moment, at most one thread may hold the lock on a mutex. Attempting to lock a mutex that is already locked either blocks or fails with an error, depending on the method used to place the lock. When a thread locks a mutex, it becomes the owner of that mutex. Only the mutex owner can unlock the mutex.

In general, we employ a different mutex for each shared resource (which may consist of multiple related variables), and each thread employs the following protocol for accessing a resource:

mỗi một resource 1 khóa mutex

- lock the mutex for the shared resource;
- access the shared resource; and
- unlock the mutex.

quy trình chuẩn để implement khóa mutex

critical section là đoạn code mà nhiều luồng cùng muốn truy cập

If multiple threads try to execute this block of code (a critical section), the fact that only one thread can hold the mutex (the others remain blocked) means that only one thread at a time can enter the block, as illustrated in below figure:

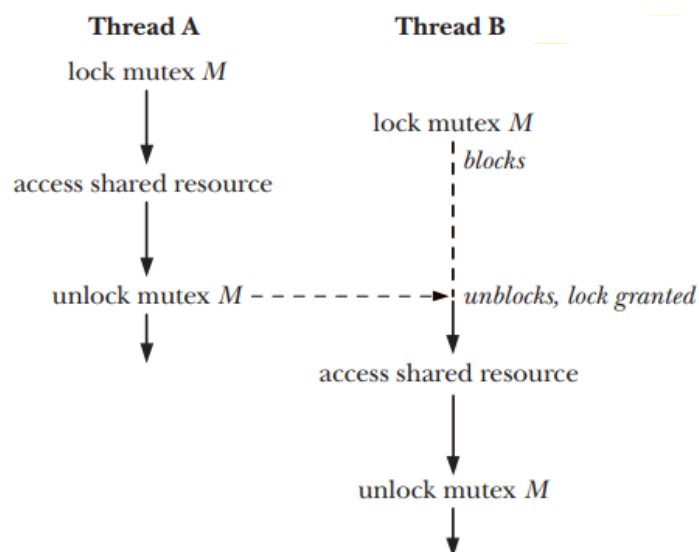


Figure 30-2: Using a mutex to protect a critical section

2.2 Allocated mutexes

A mutex is a variable of the type `pthread_mutex_t`. Before it can be used, a mutex must always be initialized. A mutex can either be allocated as a static variable or be created dynamically.

Statically Allocated Mutexes

A statically allocated mutex is declared as a global or static variable and is initialized at compile time. It is initialized using `PTHREAD_MUTEX_INITIALIZER` at compile time and available for the lifetime of the program.

Example:

```
#include <pthread.h> #include <stdio.h>

pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER; // Statically
allocated
```

Dynamically Allocated Mutex

A dynamically allocated mutex is created at runtime using `malloc()` and explicitly initialized using `pthread_mutex_init()` (The static initializer value `PTHREAD_MUTEX_INITIALIZER` can be used only for initializing a statically allocated mutex with default attributes).

```
#include <pthread.h>

int pthread_mutex_init(pthread_mutex_t *mutex, const pthread_mutexattr_t
*attr);

Returns 0 on success, or a positive error number on error
```

The mutex argument identifies the mutex to be initialized. The `attr` argument is a pointer to a `pthread_mutexattr_t` object that has previously been initialized to define the attributes for the mutex. If `attr` is specified as `NULL`, then the mutex is assigned various default attributes.

2.3 Locking and unlocking a mutex

After initialization, a mutex is unlocked. To lock and unlock a mutex, we use the `pthread_mutex_lock()` and `pthread_mutex_unlock()` functions.

nghĩa là mutex đó được khởi tạo ngay trong quá trình biên dịch chương trình, chứ không phải lúc chương trình đang chạy

```
#include <pthread.h>
```

```
int pthread_mutex_lock (pthread_mutex_t *mutex);
```

```
int pthread_mutex_unlock (pthread_mutex_t *mutex);
```

Both return 0 on success, or a positive error number on error

tại 1
thời
điểm, 1
mutex
chỉ
được
lock bởi
1 thread

To lock a mutex, we specify the mutex in a call to `pthread_mutex_lock()`. If the mutex is currently unlocked, this call locks the mutex and returns immediately.

If the mutex is currently locked by another thread, then `pthread_mutex_lock()` blocks until the mutex is unlocked, at which point it locks the mutex and returns.

If the calling thread itself has already locked the mutex given to `pthread_mutex_lock()`, then, for the default type of mutex, one of two implementation defined possibilities may result: the thread **deadlocks**, blocked trying to lock a mutex that it already owns, or the call fails, returning the error

bị block vĩnh viễn

hoặc trả
về mã lỗi

EDEADLK. On Linux, the thread deadlocks by default. Linux thì mặc định là deadlock

The `pthread_mutex_unlock()` function unlocks a mutex previously locked by the calling thread. It is an error to unlock a mutex that is not currently locked, or to unlock a mutex that is locked by another thread. (When a thread locks a mutex, it becomes the owner of that mutex. Only the mutex owner can unlock the mutex.) If more than one other thread is waiting to acquire the mutex unlocked by a call to `pthread_mutex_unlock()`, it is indeterminate which thread will succeed in acquiring it.

Nếu như có nhiều hơn 1 thread đang chờ để chiếm cái mutex đang khóa thì không xác định được thread nào sẽ chiếm được

Các yếu tố thực tế có thể ảnh hưởng:

Scheduler của kernel: Linux scheduler quyết định thread nào được chạy tiếp.

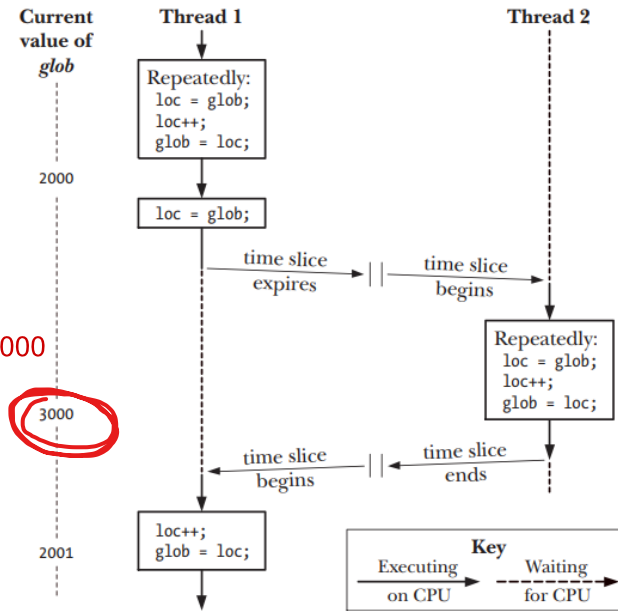
Loại mutex: Một số loại mutex (ví dụ trong RTOS) có thể hỗ trợ priority inheritance hoặc chọn thread có priority cao.

Ngẫu nhiên / nondeterministic: Nếu nhiều thread có cùng mức ưu tiên, hệ thống có thể chọn bất kỳ thread nào.

Implementation detail: Trên Linux với pthreads, thường là thread nào đang chờ sẽ được đánh thức, nhưng không đảm bảo thứ tự chính xác.

Thread 1 bắt đầu chạy, đọc glob (giả sử giá trị hiện tại là 2000). Nó lưu vào loc = 2000.
 Trước khi kịp tăng loc và ghi lại, time slice của Thread 1 hết → CPU chuyển sang Thread 2.
 Thread 2 cũng đọc glob (vẫn là 2000, vì Thread 1 chưa kịp ghi). Nó tăng loc lên 2001 và ghi glob = 2001.
 Sau đó CPU quay lại Thread 1. Thread 1 tiếp tục phần còn lại: tăng loc (vốn vẫn là 2000) lên 2001, rồi ghi glob = 2001.
 Tuần tự đúng thì phải là 2002

khả năng tài liệu nhầm, phải là 2000



race condition

Figure 30-1: Two threads incrementing a global variable without synchronization

Example:

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>

static int glob = 0;
static pthread_mutex_t mtx = PTHREAD_MUTEX_INITIALIZER;
static void * /* Loop 'arg' times incrementing 'glob' */
threadFunc(void *arg) {
    int loops = *((int *) arg);
    int loc, j, s;
    for (j = 0; j < loops; j++) {
        s = pthread_mutex_lock(&mtx);
        if (s != 0)
            errExitEN(s, "pthread_mutex_lock");
        loc = glob;
        loc++;
        glob = loc;
    }
}
```

```

s = pthread_mutex_unlock(&mtx);
if (s != 0)
errExitEN(s, "pthread_mutex_unlock");
}
return NULL; }

int main(int argc, char *argv[]) {
pthread_t t1, t2;
int loops, s;
loops = 10000000;
s = pthread_create(&t1, NULL, threadFunc, &loops);
if (s != 0)
errExitEN(s, "pthread_create");
s = pthread_create(&t2, NULL, threadFunc, &loops);
if (s != 0)

errExitEN(s, "pthread_create");
s = pthread_join(t1, NULL);
if (s != 0)
errExitEN(s, "pthread_join");
s = pthread_join(t2, NULL);
if (s != 0)
errExitEN(s, "pthread_join");
printf("glob = %d\n", glob);
exit(EXIT_SUCCESS);
}

```

2.4 Destroying mutexes

When an automatically or dynamically allocated mutex is no longer required, it should be destroyed using `pthread_mutex_destroy()`. (It is not necessary to call `pthread_mutex_destroy()` on a mutex that was statically initialized using `PTHREAD_MUTEX_INITIALIZER`.)

```
#include <pthread.h>
```

```
int pthread_mutex_destroy (pthread_mutex_t *mutex);
```

Returns 0 on success, or a positive error number on error

an toàn nhất để destroy là khi không lock và không có thread nào chờ lock nữa

It is safe to destroy a mutex only when it is unlocked, and no thread will subsequently try to lock it. If the mutex resides in a region of dynamically allocated memory, then it should be destroyed before freeing that memory region. An automatically allocated mutex should be destroyed before its host function returns. tức là phải destroy mutex bên trong hàm khởi tạo mutex đó trước khi hàm đó return

heap

A mutex that has been destroyed with `pthread_mutex_destroy()` can subsequently be reinitialized by `pthread_mutex_init()`. destroy rồi có thể khởi tạo lại

3. Condition variables

A mutex prevents multiple threads from accessing a shared variable at the same time. A condition variable allows one thread to inform other threads about changes in the state of a shared variable (or other shared resource) and allows the other threads to wait (block) for such notification.

A condition variable is always used in conjunction with a mutex. The mutex provides mutual exclusion for accessing the shared variable, while the condition variable is used to signal changes in the variable's state.

3.1 Allocated Condition Variables

A condition variable has the type `pthread_cond_t`. A condition variable must be initialized before use. cũng khởi tạo tĩnh và động như mutex

Statically allocated condition variables

A statically allocated condition variable is declared as a global or static variable and initialized at compile time. A static condition variable can be initialized using the `PTHREAD_COND_INITIALIZER` macro:

```
pthread_cond_t cond = PTHREAD_COND_INITIALIZER;
```

compile time

This ensures that the condition variable is ready to use without requiring explicit initialization.

Dynamically allocated condition variables

A dynamically allocated condition variable is created at runtime using `malloc()` and explicitly initialized with `pthread_cond_init()`. Using

Bình thường thì thread nào được cấp CPU thì chạy, nhưng chúng không có cách nào tự báo cho nhau vì thế mới đẻ ra condition variables

`pthread_cond_init()` we can initialize automatically and dynamically allocated condition variables, and to initialize a statically allocated condition variable with attributes other than the defaults.

```
#include <pthread.h>
```

```
int pthread_cond_init(pthread_cond_t *cond, const pthread_condattr_t *attr);
```

Returns 0 on success, or a positive error number on error

The `cond` argument identifies the condition variable to be initialized. As with mutexes, we can specify an `attr` argument that has been previously initialized to determine attributes for the condition variable. Various Pthreads functions can be used to initialize the attributes in the `pthread_condattr_t` object pointed to by `attr`. If `attr` is NULL, a default set of attributes is assigned to the condition variable.

3.2 Signaling and Waiting on Condition Variables

The principal condition variable operations are **signal** and **wait**. The **signal** operation is a notification to one or more waiting threads that a shared variable's state has changed. The **wait** operation is the means of blocking until such a notification is received.

Signaling a Condition Variable

A thread signals a condition variable when it changes the state of a shared resource and wants to notify waiting threads.

```
int pthread_cond_signal (pthread_cond_t *cond);
```

```
int pthread_cond_broadcast (pthread_cond_t *cond);
```

Parameter:

- `cond`: The condition variable to wait on.

Return value: return 0 on success, or a positive error number on error

The difference between `pthread_cond_signal()` and `pthread_cond_broadcast()` lies in what happens if multiple threads are blocked (wait) in

`pthread_cond_wait()`. With `pthread_cond_signal()`, we are simply guaranteed that at least one of the blocked threads is woken up; with

`pthread_cond_broadcast()`, all blocked threads are woken up.

Waiting on a condition variable → wake all threads

unlock
mutex
and
sleep

A thread **waits** on a condition variable **when it needs to be notified before proceeding**. While waiting:

- The **mutex is automatically unlocked** so **other threads can access and modify the shared resource**. khi thread A đang wait thì các thread khác có quyền truy cập
- The **thread is put into a blocked state (sleeps)** until another thread signals it. thread A sẽ sleep cho đến khi có signal từ thread khác đến
- **Once signaled, it relocks the mutex and continuing execution**, thread then **immediately accesses the shared variable**.

The `pthread_cond_wait()` function automatically **performs the mutex unlocking and locking**.

```
#include <pthread.h>
int pthread_cond_wait(pthread_cond_t *cond, pthread_mutex_t *mutex);
```

Parameter:

- cond: The condition variable to wait on.
- mutex: The mutex protecting the shared resource.

Return value: return 0 on success, or a positive error number on error

Example:

```
#include<stdio.h>
#include<pthread.h>
pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t cond = PTHREAD_COND_INITIALIZER;
int ready = 0; // Shared condition variable

void *prodfun(void *arg)
{
    pthread_mutex_lock(&lock);
    ready = 1;
    printf("Producer: Ready, signal consumer\n");
    pthread_mutex_unlock(&lock);
    pthread_cond_signal(&cond);
}

void *consfun(void *arg){
    pthread_mutex_lock(&lock);
    while (!ready){
        printf("Consumer: Waiting...\n"); pthread_cond_wait(&cond, &lock);
    }
}
```

```

}
printf("Consumer: Data received.\n");
pthread_mutex_unlock(&lock);
}

void main()
{pthread_t prod, cons;
pthread_create(&prod, NULL, prodfun, NULL);
pthread_create(&cons, NULL, consfun, NULL);
pthread_join(prod, NULL);
pthread_join(cons, NULL);
pthread_mutex_destroy(&lock);
pthread_cond_destroy(&cond);
}

```

3.3 Destroying condition variable

When an automatically or dynamically allocated condition variable is no longer required, then it should be destroyed using *pthread_cond_destroy()*. It is not necessary to call *pthread_cond_destroy()* on a condition variable that was statically initialized using *PTHREAD_COND_INITIALIZER*.

```

#include <pthread.h>

int pthread_cond_destroy(pthread_cond_t *cond);

Returns 0 on success, or a positive error number on error

```

It is safe to destroy a condition variable only when no threads are waiting on it. If the condition variable resides in a region of dynamically allocated memory, then it should be destroyed before freeing that memory region. An automatically allocated condition variable should be destroyed before its host function returns. A condition variable that has been destroyed with *pthread_cond_destroy()* can subsequently be reinitialized by *pthread_cond_init()*.